Experiment 6

Dashboard and Scorecard Development using Microsoft Power BI

Business Intelligence Lab · Source file: exp_6.pbix · Dataset: Global Superstore

$12.64M	$1.47M	25,728	11.6%
Total Sales	Total Profit	Total Orders	Profit Margin

Aim

To design an interactive executive dashboard and a business scorecard for Global Superstore that monitor sales, profit, orders and margin, and to turn the findings into a recommendation for management.

Headline findings

• **Best region:** Western Europe, with the highest sales ($1,731,893) and the highest profit ($218,423).
• **Worst region:** Western Asia, a loss of $53,922 on $317,107 of sales (margin -17.0%).
• **Most profitable category:** Technology ($663,768). **Least profitable:** Furniture ($285,016, margin 6.9%).
• **Recommendation:** review the pricing and discounting of Tables, the only loss-making Furniture sub-category (–$64,083).

How to read this report

The images in this report are charts rebuilt from the data stored in exp_6.pbix, laid out the way the dashboard is specified in the lab sheet. They are not screenshots of Power BI. The saved file contains only part of the dashboard (see section 7), so sections 3 to 5 show the complete dashboard the brief asks for, and section 3.1 shows what is actually saved.

1. Dashboard planning (Part A)

Management needs to answer four questions at a glance: how much are we selling, are we profitable, how many orders are we handling, and where and from whom is the performance coming. That gives four headline KPIs and five supporting charts, each chosen for the question it answers.

Metric / visual	Type	Question it answers
Total Sales, Total Profit, Total Orders, Profit Margin %	KPI cards	How is the business doing overall?
Sales by Region	Bar chart	Which markets drive revenue?
Sales by Category	Column chart	What do we sell most?
Monthly Sales Trend	Line chart	Is the business growing; is there seasonality?
Top 10 Customers by Sales	Bar chart	Who are our key accounts?
Profit by Segment	Column chart	Which customer segment earns the most?
Region, Category, Year	Slicers	Let each manager focus on their own view.
Target vs Actual (Sales, Profit, Orders)	Scorecard table with colour status	Are we meeting our goals?

Layout sketch



Slicers and title run across the top, KPI cards sit directly below so they are read first, and charts and the scorecard fill the lower two rows.

2. KPI dashboard (Part B)

The finished one-page dashboard with all slicers set to “All”. The KPI cards use the same definitions as the model: Total Sales = SUM of Sales, Total Profit = SUM of Profit, Total Orders = distinct count of Order ID, and Profit Margin = Profit ÷ Sales.

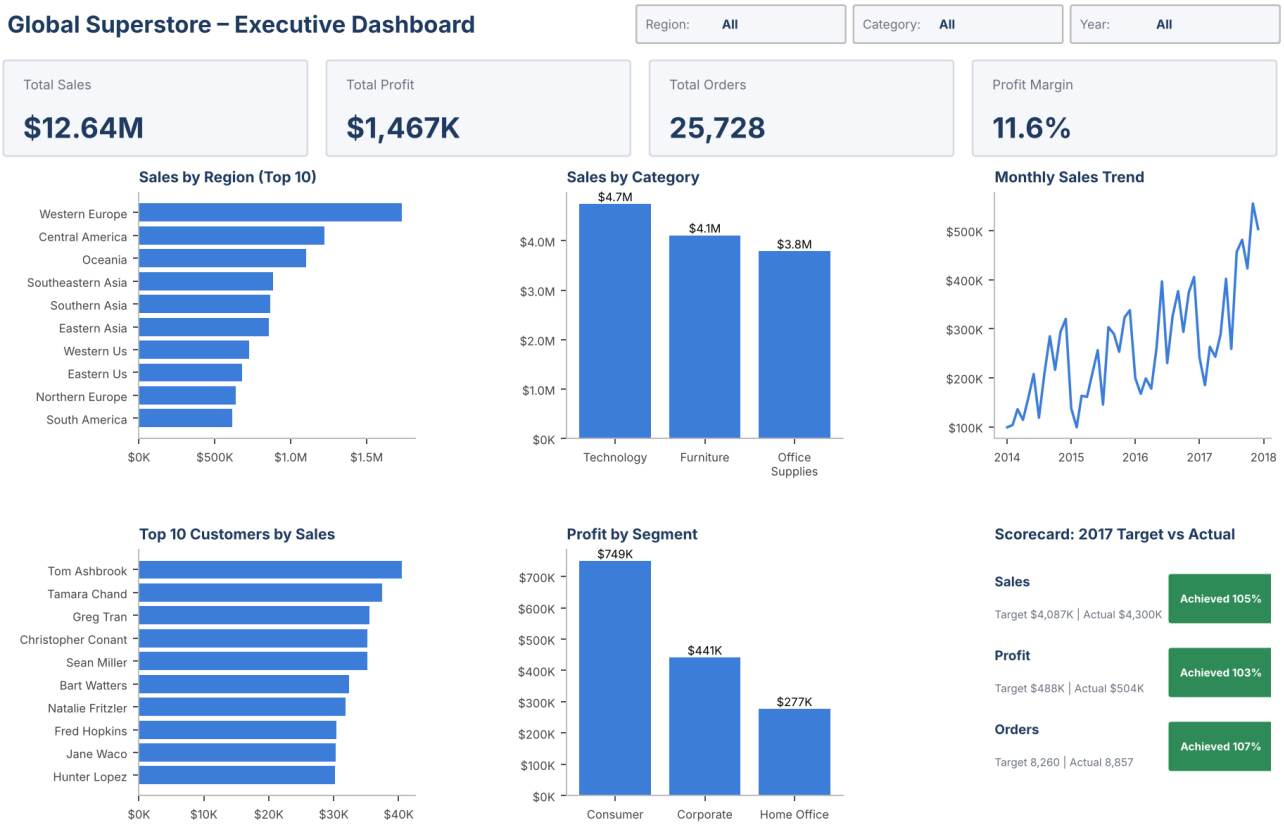


Figure 1. Executive Dashboard, all data.

Formatting applied for readability: one blue colour for ordinary values and red only for losses, money axes shown in $K/$M, titles on every visual, and the same fonts and card style throughout.

3. Interactivity (Part C)

Three slicers (Region, Category, Year) filter every visual and KPI card on the page at the same time. The figure below shows the dashboard with **Region = Western Europe** and **Year = 2017** selected: every card and chart changes, which is how the slicer test (step 9) is checked.

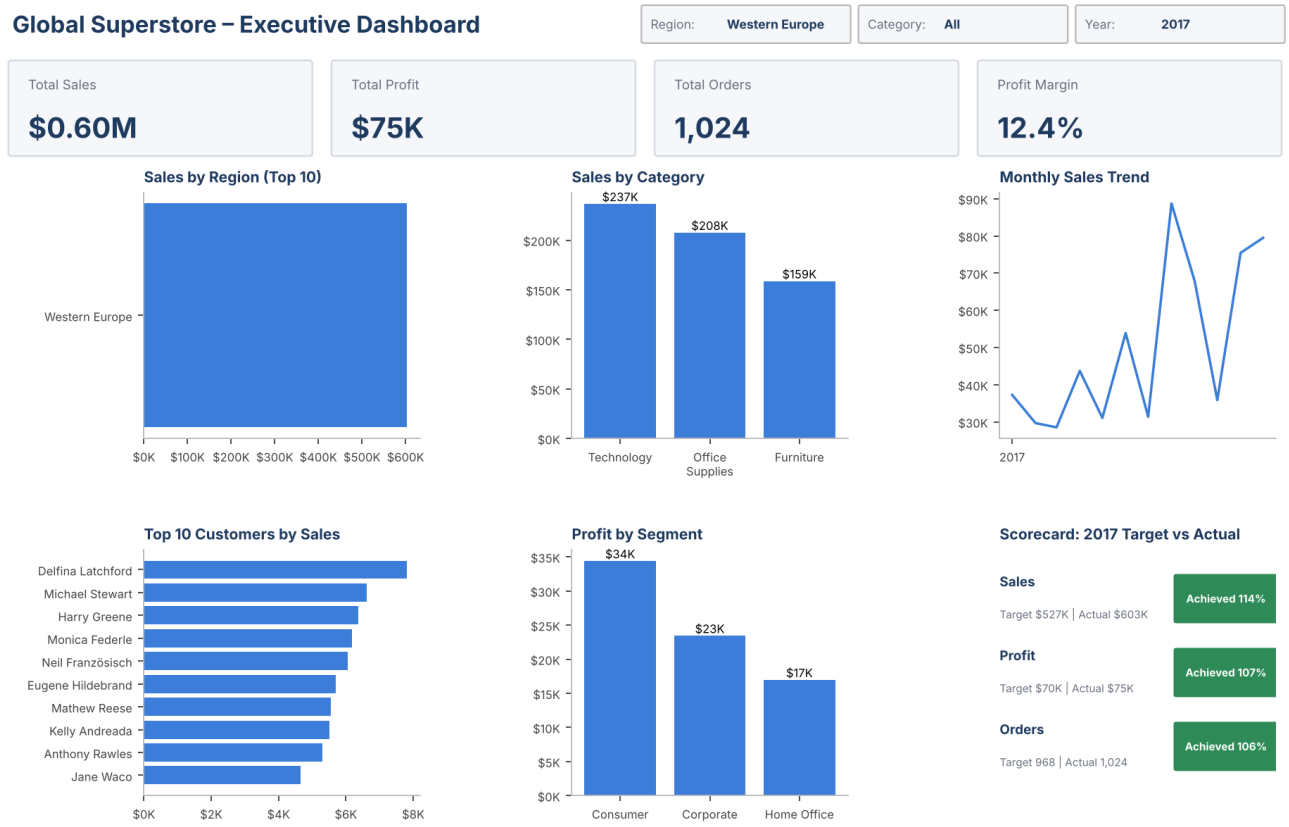


Figure 2. Dashboard filtered to Western Europe, 2017.

Test result. With the filter on, the cards read Sales $602,861, Profit $74,775, Orders 1,024 and Margin 12.4%; each is a subset of the all-data totals, and the top customers and monthly trend change to match. The dashboard is therefore filtering as expected.

3.1 What is actually saved in exp_6.pbix

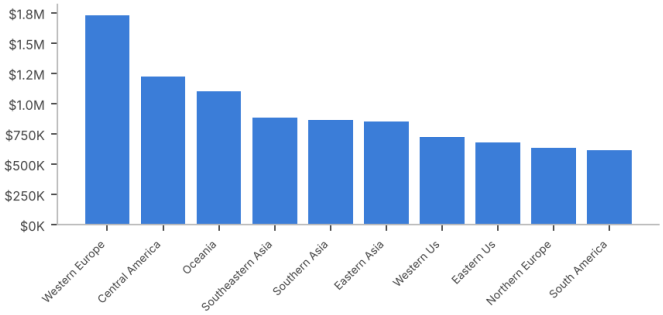
The file's **Executive Dashboard** page has four visuals and no cards or slicers. They are rebuilt below from the model data.

Executive Dashboard page – visuals as saved in exp_6.pbix (rebuilt from model data)

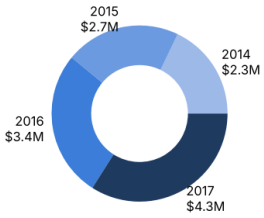
Table: Sales, Profit, Order ID, Category

Order ID	Category	Sales	Profit
CA-2014-SM20320140-	Technology	23,460	-1,828
CA-2016-TC20980140-4	Technology	17,500	8,400
CA-2017-RB19360140-4	Technology	14,000	6,720
CA-2017-TA21385140-4	Technology	13,653	4,568
CA-2017-HL15040140-4	Technology	10,500	5,040
CA-2014-SC20095140-4	Office Suppl	9,900	4,669
CA-2016-AB10105140-4	Office Suppl	9,893	4,946
US-2016-BS11365140-4	Technology	9,100	2,366

Sales by Region (Top 10)



Donut: Sales by Year (drill: Year > Quarter > Month > Day)



Line: Sales and Profit by Region

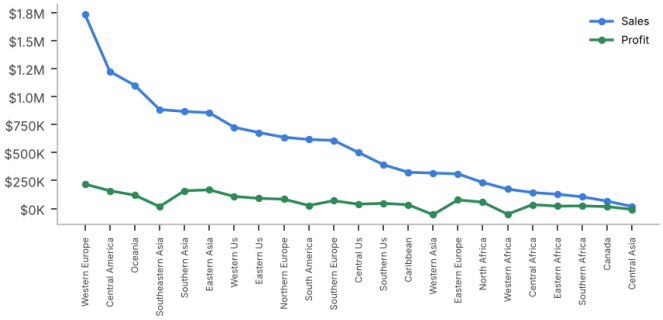


Figure 3. Visuals on the saved Executive Dashboard page.

The file also has five working pages from earlier experiments (exp.3, button slice, slicer, list slice, input slicer) that test slicer styles on a Region/City/Country hierarchy. They show the slicers work on this model, but they are not on the dashboard page.

4. Business scorecard (Part D)

The scorecard compares Actual 2017 results with a Target. The file contains no targets, so the targets here are **illustrative**: each is 20% above the 2016 actual. Replace them with management's real targets. Status rule: 100% or more = **Achieved**; 80–99% = **Partially Achieved**; below 80% = **Not Achieved**. In Power BI the Status column is coloured by conditional formatting on the same rule.

Measure (2017)	Target	Actual	Achievement	Status
Sales	$4,086,742	$4,299,763	105.2%	Achieved
Profit	$488,267	$504,219	103.3%	Achieved
Orders	8,260	8,857	107.2%	Achieved

Figure 4. Scorecard, company level.

By category

Category	Measure	Target (2016 +20%)	Actual 2017	Achievement	Status
Furniture	Sales	$1,341,155	$1,377,902	102.7%	Achieved
Furniture	Profit	$100,857	$89,287	88.5%	Partially Achieved
Office Supplies	Sales	$1,212,821	$1,305,759	107.7%	Achieved
Office Supplies	Profit	$179,059	$180,014	100.5%	Achieved
Technology	Sales	$1,532,766	$1,616,102	105.4%	Achieved
Technology	Profit	$208,352	$234,917	112.8%	Achieved

Reading the scorecard. At company level all three targets are met. By category, Furniture profit reaches only 88.5% of its target, so it is **Partially Achieved** even though Furniture sales are on track. That matches the margin problem found in the analysis.

5. Business analysis (Part E)

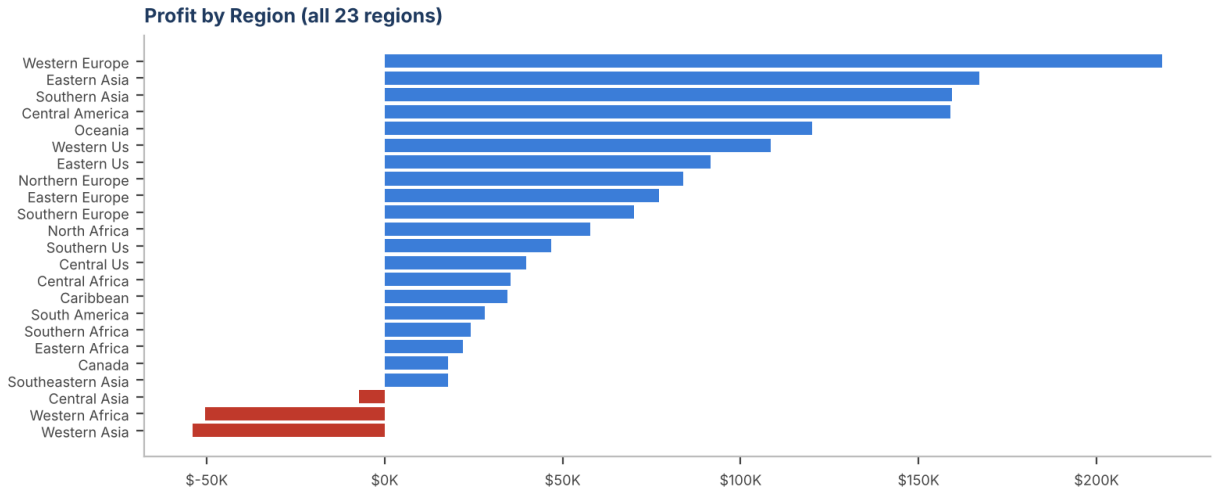
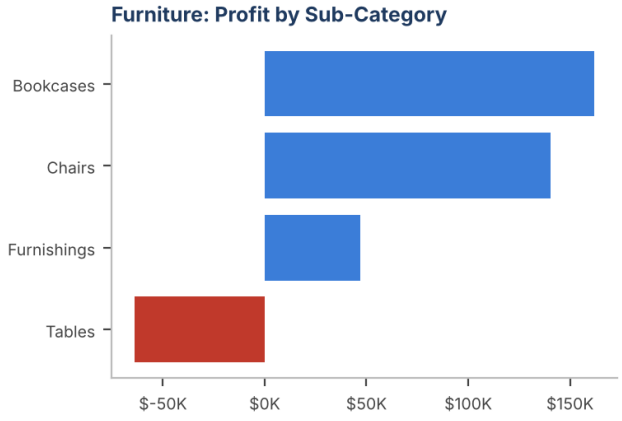


Figure 5. Profit by region. Red bars are loss-making.

Question	Answer	Evidence
Best performing region	Western Europe	Highest sales ($1,731,893) and highest profit ($218,423); margin 12.6%.
Worst performing region	Western Asia	Largest loss: $-53,922 on $317,107 sales (-17.0%). Western Africa ($-50,408) and Central Asia ($-7,282) also lose money.
Most profitable category	Technology	$663,768 profit on $4,744,501 sales (14.0% margin).
Least profitable category	Furniture	$285,016 profit on $4,109,197 sales (6.9% margin, half that of the other categories).

Furniture drilled down



Furniture sells almost as much as Technology but earns half the margin. The cause is one sub-category: **Tables** sold $757,042 and lost $64,083, while Bookcases, Chairs and Furnishings all make a profit.

Business recommendation. Review the pricing, discounting and supplier cost of Tables, starting with the loss-making regions (Western Asia, Western Africa, Central Asia). Tables alone cost the company $64K of profit; bringing them to break-even would lift the Furniture margin from 6.9% to about 8.5% and close most of the Furniture profit gap in the scorecard.

6. Result

An executive dashboard with KPI cards, five charts and three slicers, and a target-versus-actual scorecard with status indicators, were designed on the Global Superstore model. Analysis shows Western Europe and Technology lead, Western Asia and Furniture lag, and Tables are the main source of lost profit.

7. Check against the brief and the file

“Done” is in the saved file, “Partly” is started, “No” is missing from the file (shown in this report only).

Brief item	Status	Detail
KPI cards: Sales, Profit, Orders, Margin %	No	Not on the Executive Dashboard page. Cards exist only on the exp.3 page, for Sales, Profit, Quantity and Discount. There is no Orders or Margin card.
Five required charts	Partly	Saved page has Sales by Region (Top 10). Missing: Sales by Category, Monthly Sales Trend, Top 10 Customers, Profit by Segment. A donut by Year/Quarter/Month/Day and a region line chart are present but not requested.
Slicers: Region, Category, Year	No	Slicers exist on other pages (Region/City/Country) but none for Category or Year, and none on the dashboard page.
Scorecard with targets	No	No target measures, scorecard visual or conditional formatting in the file.
Business analysis and recommendation	Done	Completed in section 5 of this report.

Model points to fix before adding the missing visuals

• **Dim Customer** has 51,290 rows for 17,415 customers and joins to the fact table many-to-many. Remove duplicates so Customer ID is unique and the link becomes many-to-one.
• **Dim region** links on Region only (many-to-many, both directions). That is fine for region totals but wrong below region level.
• **Discount** is stored as a whole number, so fractional discounts became 0 or 1. Reload it as a decimal.
• Fact and Orders tables are mixed on some pages, and measures such as Total Sales and Profit Margin from the earlier model are no longer in the file. Recreate them as DAX measures (Total Sales, Total Profit, Total Orders = DISTINCTCOUNT of Order ID, Profit Margin = DIVIDE of the two).
• Rename the pages (“exp.3”, “input slicer” and so on) to meaningful names.